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High-potential control for durability improvement of the vehicle fuel cell system based on oxygen partial pressure regulation under low-load conditions

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HIGHLIGHTS

- Originally explore dynamic and control issues of cathode recirculation in vehicle PEMFC system.
- A novel control model of PEMFC system with cathode recirculation is established and validated by experiments.
- Active control of the oxygen partial pressure is achieved through a cathode recirculation pump.
- Fuzzy logic piecewise PID controller with feedforward compensation obtains the optimal high-potential control effect.

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ABSTRACT

In the state-of-the-art high-power self-humidifying proton exchange membrane fuel cell (PEMFC) systems for vehicles, the high potential and low water production at idle or low load conditions strongly cause corrosion and decay of key materials and thus reduce durability. Therefore, the control technology of system-level durability requires an innovative design. Cathode recirculation is beneficial in alleviating the above unfavorable factors from the perspective of regulating oxygen and vapor partial pressure. This paper presents a pioneering study on the dynamics and control of cathode recirculation in vehicle high-power self-humidifying PEMFC system under low load conditions. First, a control-oriented dynamic model of the vehicle PEMFC system with a cathode recirculation loop is developed and the steady-state and dynamic performance is verified with experimental data from a 120 kW system. Active control of the intake component is achieved by re-feeding the reacted cathode gas to the air compressor outlet through a recirculation pump. On this basis, a high-potential controller based on oxygen partial pressure regulation is designed in combination with the dynamics of cathode recirculation. Results show that the designed dynamic fuzzy logic segmented proportional integral derivative controller with feedforward compensation achieves the optimal high-potential control effect by managing the oxygen partial pressure under variable low load conditions. It not only has excellent anti-disturbance ability but also effectively reduces the dynamic response time, transient overshoot, and steady-state error to satisfy the rapid and stable voltage output. Finally, the concomitant effect of humidification brought by the implementation of the optimal high-potential controller is analyzed, and the results show that the proton membrane is completely humidified.

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Nomenclature

Variables

v	voltage
T	temperature
J	inertia
t	time
p	pressure
n	rotating speed or amount of substance
W	mass flow
R	ideal gas constant or resistance
M	molar mass
C_D	discharge coefficient
D	diameter
V	volume
H	enthalpy
h	specific enthalpy
I	current
C	concentration
i	current density
A_{cell}	active area of cell
N	cell number
F	faraday constant
n_d	electroosmotic drag coefficient
D_w	water diffusivity coefficient
l	thickness
a	activity
e	error
A	ampere
i_{max}	limiting current density

Abbreviations

UTC	united technologies corporation
PEMFC	proton exchange membrane fuel cell
RBF	radial basis function
BP	back-propagation
BPV	back pressure valve
RH	relative humidity
OER	oxygen excess ratio
CCP	cathode circulation pump
SFF	feedforward
PID	proportional-integral-derivative
SPID	segment pid
FSPID	fuzzy logic segment pid
IAC	integral absolute cost
ITAC	integral time-weighted absolute cost

Symbol

ω	angular velocity
τ	torque or time constant
η	efficiency
γ	isentropic coefficient
π	pi
θ	angle
λ	membrane water content
ξ	Proportion coefficient
ρ	density

Subscript

cp	compressor
cm	compressor motor
sm	supply manifold
atm	atmosphere
ccp	cathode circulation pump
rm	return manifold
out	outlet
in	input
ca	cathode
an	anode
mix	mix gas
O_2	oxygen
N_2	nitrogen
v	water vapor
sat	saturation
act	activation
ohm	ohmic
conc	concentration
m	membrane

Introduction

Proton exchange membrane (PEM) fuel cell as a power supply device has the advantages of high efficiency, high power density, no pollution and low noise [1,2]. Therefore, under the wave of global new energy vehicle transformation, fuel cell systems consisting of several hundred individual PEMFCs have become the main candidates for automotive power systems and have made great progress on the road to commercialization through technology accumulation [3,4]. However, in the current state-of-the-art automotive fuel cell systems, the durability issue is still a huge challenge due to the widely varying operating conditions and complex and variable environmental factors [5–8].

The durability degradation problem of automotive fuel cell systems can be almost contained in the five most common operating conditions: low load (idle), variable load, start-stop, high load, and low temperature start [8,9]. The current durability optimization control techniques for automotive fuel cell systems are also focused on the above five operating conditions [10–12]. Specifically, under low load or idling conditions, the system suffers from a small current dynamic, which exposes the PEMFC to a high-potential cycle and a dehydration atmosphere for a long time [13]. Previous studies have shown that high-potential cycling can significantly cause carbon corrosion and platinum aggregation, resulting in irreversible loss of electrochemical surface area [14–16], meanwhile, water shortage causes membrane dehydration and increases the risk of membrane rupture [17,18]. In particular, the principle of the self-humidification function used in the current state-of-the-art fuel cell system strongly relies on the amount of water produced by the cell itself, and the low water production volume under low load (idle) conditions inevitably aggravates the damage to the proton membrane of the self-humidification system [19]. The consequences of material degradation due to high potential and water deficiency problems under low load (idle) conditions cause an irreversible decrease in system durability. Specifically, it is noted that low load (idle) conditions occupy a great portion of the whole vehicle fuel cell system life cycle [20]. Furthermore, in the current research and development of high-power fuel cell systems for vehicles, the general control strategy for idle conditions is to increase the idle current to limit the output voltage below 0.8v in order to avoid excessive material attenuation caused by high potential. Under this idle control strategy, the idle power of the system is much higher than the parasitic power of all the auxiliary components (compressor, hydrogen circulation pump, water pump, etc.), which results in a waste of power. The ideal idle operation mode is that the stack power is equal to the parasitic power, which requires a reduction in idle current, thus pushing back the voltage to a higher level. This is a major contradiction in the current idling conditions of vehicle fuel cell systems. Therefore, innovative designs are urgently needed to improve system durability under low load (idle) conditions.

The cathode recirculation scheme is able to mitigate the durability degradation caused by low load (idling) conditions theoretically [21]. It is similar in organization to anode recirculation. The principle is that controlling a recirculation device re-feeds gas with high humidity and low oxygen concentration in the cathode outlet back to the cathode inlet and mixes it with fresh air from the air compressor. The oxygen partial pressure and vapor partial pressure of the mixed inlet air decreased and increased respectively thus having the effect of limiting the high potential and increasing the humidity [22]. When high potential control based on oxygen partial pressure regulation is achieved by using cathode recirculation, which increases the possibility to control the voltage at a lower level and reduces the idling current, thus reducing the idling power, delaying material decay, and improving durability.

At present, a great many studies have discussed the potential of using cathode recirculation to regulate fuel cell inlet gas components (nitrogen, oxygen, and water vapor) and thus achieve limiting high potential and humidification effects in fuel cells. UTC power [23] pioneered the use of cathode exhaust gas to reduce the open-circuit voltage of fuel cells to alleviate decay, however, this is only a conceptual contribution and does not consider the application scenarios of automotive fuel cell systems. Similarly, Hydrogenics corporation [24] proposed the use of cathode recirculation to adjust the polarization curve of the fuel cell stack in their patent, but these works were not practically implemented. Cheng et al. [25] designed a PI controller to limit the high potential through a simplified simulation model based on the principle of oxygen concentration regulation by cathode recirculation, and the results showed the high potential suppression effect. However, the performance of the designed controller was only investigated at a fixed current. On this basis, Jiang et al. [26] conducted a series of dynamic experimental studies on cathode and anode dual circulation on a stack test bench, which showed that 98% of the inlet gas relative humidity could be achieved by using cathode recirculation, and voltage clamping could also be achieved using anode recirculation due to the reduction of hydrogen partial pressure, but the effect was much lower than that of cathode recirculation. Meanwhile, Zhao et al. [27] analyzed the steady-state voltage clamp and humidification capability of cathode-anode double-cycling on a 10 kW system through an orthogonal experimental method and obtained qualitative conclusions similar to Jiang's. Kim et al. [28] investigated the cathode recirculation of fuel cells and verified the self-humidification effect by theoretical and experimental methods. Tests in a 5-cell stack showed that the humidification performance of cathode recirculation was slightly lower than that of a membrane humidifier. Shao et al. [29] compared and analyzed the difference in humidification performance between cathode and anode recirculation and showed that the humidification capacity of cathode recirculation was much higher than that of anode recirculation especially at low inlet humidity, but the power consumption of cathode recirculation was much higher than that of anode recirculation. Xu et al. [30] developed a detailed control-oriented dynamic model of a fuel cell system to simulate the water management capabilities of cathode recirculation, and the simulation results showed that parameters such as air stoichiometry, current density, cathode recirculation ratio, and membrane thickness had a significant impact on system performance. Through precise control of the cathode recirculation ratio and careful selection of design and operating parameters, a system efficiency similar to that of a PEMFC equipped with an external humidifier can be achieved. Zhang et al. [31] explored the humidification and voltage limiting effects of cathode recirculation by means of simulation plus experiments, respectively. The high-frequency impedance results showed that the introduction of cathode recirculation significantly reduced the impedance value of the membrane.

The aforementioned important studies have contributed positively to the application of cathode recirculation in fuel cells and in particular have qualitatively demonstrated that cathode recirculation achieves high potential suppression or humidification by changing the partial pressure of oxygen and water vapor. However, there are still many knowledge gaps to be bridged for the application of this important research orientation to fuel cells. Especially, the issues of dynamic and control become prominent in the changing low load conditions when cathode recirculation is expected to be applied to automotive high-power fuel cell systems to develop optimal control techniques for durability under low load conditions. Based on the above arguments, this paper further improves and broadens the work: (i) While previous applications of cathode recirculation have been in small-scale reactors or systems, this paper extends it further to high-power vehicle self-humidifying fuel cell systems. (ii) In modeling systems with cathode recirculation, previous studies have generally directed the recirculation loop back to the compressor inlet through a solenoid valve. This approach is suitable for studying the operating mechanism but is not applicable to automotive systems since it would expose the compressor to long-term harsh operating conditions. In this paper, a recirculation pump is used to transport the gas to the compressor outlet for mixing, avoiding damage to the compressor and facilitating its active control. (iii) The problem of dynamics and control of cathode recirculation on automotive systems is a significant gap, which is innovatively focused on in this paper. Therefore, combining the above points, the main contents and contributions of this paper are as follows:

(1) An accurate control-oriented model of a high-power self-humidifying fuel cell system equipped with a cathode recirculation pump is established, and the cathode gas supply loop and voltage model are experimentally validated, which indicates that the complex gas supply process and performance output can be accurately evaluated.
(2) The dynamic of cathode recirculation is analyzed. Based on the established nonlinear time-varying model and combined with the operating characteristics of the advanced vehicle high-power system, a high potential control strategy on account of oxygen partial pressure regulation is developed to improve the durability under low load conditions.
(3) A two-dimensional fuzzy logic controller is designed to optimize the PID algorithm for oxygen partial pressure regulation, which achieves accurate control of the high potential in terms of variable load responsiveness, overshoot, robustness, and steady-state following error. Meanwhile, the concomitant changes in humidity are analyzed under the implementation of the optimal high-potential controller to achieve synergistic optimization.
(4) The designed high potential controller can avoid the corrosion of materials by high potential on the basis of reducing the idling current, making it possible to further reduce the idling power output and avoid power wastage.
(5) Overall, this paper innovatively investigates the operation and control of cathode recirculation on automotive PEMFC systems based on the control model, which provides a new orientation for the development and implementation of the current durability control technology for automotive high-power fuel cell systems, with a positive reference value and engineering significance.

Modeling and validation of the PEMFC system with cathode recirculation

System structure and model hypothesis

The schematic and actual bench diagram of the developed automotive self-humidifying fuel cell system with cathode recirculation is shown in Fig. 1. This system consists of a stack and four essential key auxiliary systems. The stack module is the site where hydrogen and oxygen are consumed to generate water and provide electrical energy through an electrochemical reaction. The gas supply subsystem differs from the conventional one due to the addition of a cathode recirculation device, and includes key components such as a centrifugal air compressor, a backpressure valve, an intercooler, an exhaust gas recirculation pump and transmission pipelines. The modeling and control of this new gas supply subsystem is the focus of this paper. Another innovative application is the elimination of the external humidifier in favor of a self-humidification function, which is achieved by an advanced internal design of the stack and a hydrogen-air counter-flow configuration. In addition, the hydrogen subsystem, the water and thermal management subsystem, and the electrical subsystem are not different from those of a generic automotive fuel cell system. What follows in this paper is the modeling of the key components and processes of the above-mentioned automotive PEMFC system with cathode recirculation function. In order to balance the speed and accuracy of the developed system control model, the following important assumptions must be stated:

(1) All gases follow the basic laws of ideal gases, and the dynamic processes follow the laws of mass conservation and energy conservation
(2) The optimal operating temperature of the reactor under low load conditions is almost constant (Lower than 90A), so the temperature is set to 338K in the model, which is also consistent with the experiment.
(3) The performance degradation caused by flooding is neglected due to the low water production under low load conditions.
(4) The humidity of the atmosphere as well as the hydrogen source is set to a constant value.

Air compressor

Dynamic modeling

The dynamic behavior of the air compressor speed ω_{cp} can be captured by the dynamic balance equation between the motor and the compressor [32].

$$J_{cp}\frac{d\omega_{cp}}{dt} = (\tau_{cm} - \tau_{cp}) \tag{1}$$

Where the drive motor torque τ_{cm} can be described as a function of compressor input voltage and speed by the static motor equation:

$$\tau_{cm} = \eta_{cm}\frac{k_t(v_{cm} - k_v\omega_{cp})}{R_{cm}} \tag{2}$$

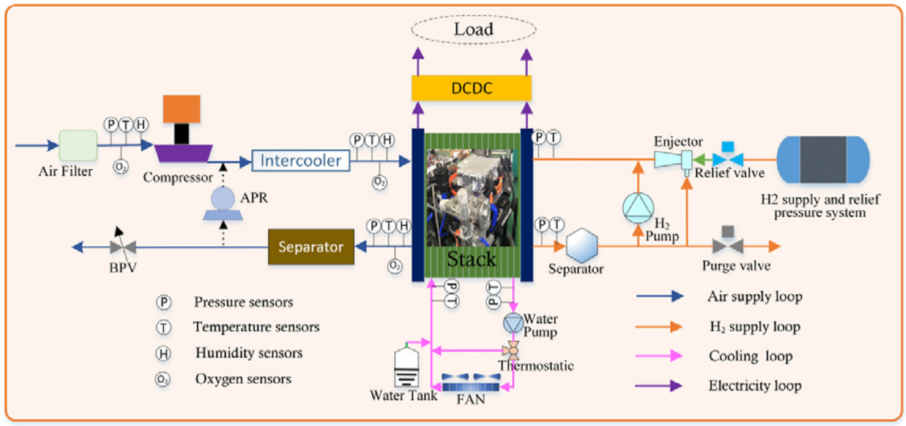
Meanwhile, the driving compressor torque τ_{cp} can be described as a function of flow rate, pressure ratio and speed by the thermodynamic equation:

$$\tau_{cp}\omega_{cp} = C_p\frac{T_{atm}}{\eta_{cp}}\left[\left(\frac{p_{sm}}{p_{atm}}\right)^{\frac{\gamma-1}{\gamma}} - 1\right]W_{cp} \tag{3}$$

The dynamic response characteristics of the air compressor can be obtained by equations (1) (2) (3), the only control variable being the input voltage v_{cm}. The compressor flow rate W_{cp} can be described as a function of speed and pressure ratio as follows.

$$W_{cp} = f\left(\omega_{cp}, \frac{P_{sm}}{P_{atm}}\right) \tag{4}$$

The air compressor outlet temperature $T_{cp,out}$ is approximated by adiabatic compression of an ideal gas as follows:



(a)

(b)

Fig. 1 – Self-humidifying fuel cell system developed for passenger cars used in this paper. (a) System structure diagram; (b) actual system used for relevant validation experiments.

$$T_{cp,out} = T_{atm} + \frac{T_{atm}}{\eta_{cp}}\left[\left(\frac{p_{sm}}{p_{atm}}\right)^{\frac{\gamma-1}{\gamma}} - 1\right] \tag{5}$$

Compressor RBF model and validation

Equation (4) states the compressor flow rate as a function of speed and pressure ratio. In this paper, a radial basis function (RBF) neural network is used to model the compressor flow and the model is validated by experimental data. The flow RBF neural network model avoids the large number of parameter identification problems associated with the nonlinear curve fitting methods of many previous papers, simplifying the model complexity and improving accuracy. The inputs of the RBF model are pressure ratio and rotational speed, and the output is flow rate. The training, validation and accuracy analysis of the model are shown in Fig. 2. Specifically, Fig. 2(a) shows the structure of the RBF neural network used and the model property information. Fig. 2(b) shows the 3D comparison between the experimental values of the collected 89 data points and the output values of the RBF model. It can be seen that the experimental values are in high agreement with the output values, proving the high accuracy of the flow RBF model. Fig. 2(c) shows the Map of the compressor flow derived from the obtained RBF model, which contains the pressure ratio, speed, and flow information for the full range of operating conditions, which will provide the basis for the subsequent compressor control. Fig. 2(d) shows the accuracy validation of the RBF model from four different metrics, and the results show that the high accuracy output of the RBF model is sufficient for control applications.

Cathode circulation pump

Dynamic modeling

The dynamic behavior of the cathode recirculation pump is modeled differently from the general lumped rotational parameter model with inertia. In this paper, considering the pump inertia, the dynamic response process of the cathode recirculation pump is equivalent to a first-order inertial link [33]:

$$n_{actual}(s) = \frac{1}{\tau_{cp}s + 1} n_{input}(s) \tag{6}$$

Where, the only input control variable for the cathode circulation pump is the rotational speed n_{input}.

The change in gas temperature before and after the circulation pump is approximated by the adiabatic compression of ideal gas as follows:

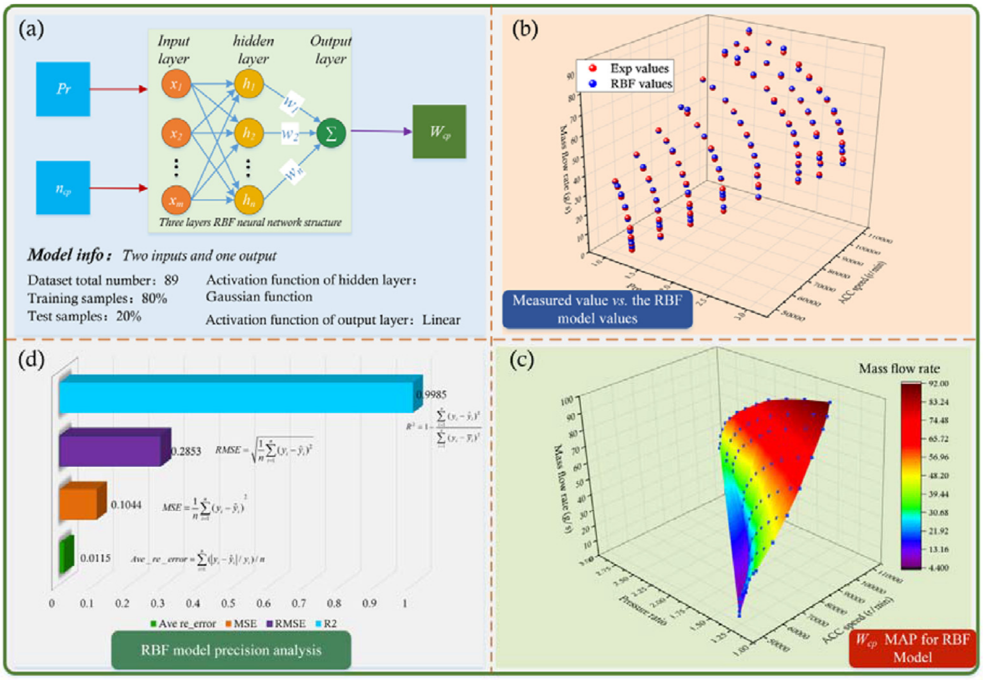


Fig. 2 – Compressor flow RBF neural network model and validation.

$$T_{ccp,out} = T_{rm} + \frac{T_{rm}}{\eta_{ccp}}\left[\left(\frac{p_{sm}}{p_{rm}}\right)^{\frac{\gamma-1}{\gamma}} - 1\right] \tag{7}$$

Flow BP model and validation

The flow MAP of the cathode recirculation pump is obtained by means of a BP artificial neural network model. Considering the operating characteristics of the recirculation pump, the flow rate W_{ccp} is defined as a function of the front-to-back pressure difference ΔP_{ccp} and the input speed n_{ccp}.

$$W_{ccp} = f(n_{ccp}, (P_{sm} - P_{rm})) \tag{8}$$

Therefore, the inputs to the BP model for circulation pump flow are differential pressure and speed, and the output is the flow rate. The attribute information of the BP ANN model is shown in Table 1.

Fig. 3 shows the comparison of the output values of the trained BP model with the experimental values. The scattered dots represent the output values of the model and the solid lines represent the experimental values. It can be seen that the developed BP model fits the experimental values very well and most of the errors are controlled within ±5%. Moreover, a randomly selected value at each speed was validated as shown in the solid gray circle, and the results also show that the validation has a high accuracy. Therefore, the developed MAP model for cathodic circulation pump flow can be satisfied for the control application.

Table 1 – BP model information for flow MAP.

Model info	Characteristics
Network type	Feed-forward backprop
Layers number	3
Hidden Neuron number	layer1:10; layer2:11
Transfer function	TANSIG
Training function	TRAINLM

Manifold dynamics model

The manifold in a fuel cell system consists of a supply manifold and a return manifold. The supply manifold contains the lumped volume from the compressor outlet to the stack inlet, and the return manifold includes the lumped volume from the stack outlet to the backpressure valve. The manifold dynamics control equations are summarized in Table 2.

Intake mixing cavity model

Since the mixing of the cathode outlet exhaust gas with the fresh air from the air compressor changes the gas composition of the inlet charge as well as the temperature and humidity, the mixing process of the two streams needs to be modeled.

The gas constants of the gas mixture can be expressed as:

$$R_{mix} = \frac{R}{M_{mix}} \tag{9}$$

The molar mass M_{mix} of the gas mixture is determined by the fresh air and the recirculated exhaust gas, and the molar mass M_{cp} of the fresh air is as follows.

$$M_{cp} = 0.21(1 - x_{v,cp})M_{O_2} + 0.79(1 - x_{v,cp})M_{N_2} + x_{v,cp}M_{H_2O} \tag{10}$$

The molar fraction of each gas (N2, O2, water vapor) in the fresh air from the air compressor can be expressed as:

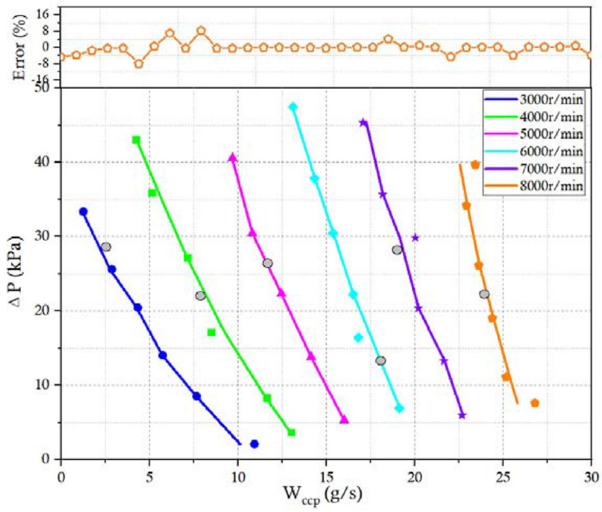


Fig. 3 – Comparison and error analysis between the output of the BP model and the experimental value. (Gray circles represent validation values, colored scatter points represent model output values, Colored solid lines represent experimental values).

$$\begin{cases} n_{O_2,cp} = 0.21(1 - x_{v,cp})\dfrac{W_{cp}}{M_{cp}} \\ n_{N_2,cp} = 0.79(1 - x_{v,cp})\dfrac{W_{cp}}{M_{cp}} \\ n_{v,cp} = (1 - x_{v,cp})\dfrac{W_{cp}}{M_{cp}} \end{cases} \tag{11}$$

Similarly, the molar mass of the recirculated gas M_{ccp} is calculated as follows.

$$M_{ccp} = (1 - x_{v,ccp})(x_{O_2,dryair,ccp} M_{O_2} + (1 - x_{O_2,dryair,ccp})M_{N_2}) + x_{v,ccp} M_{H_2O} \tag{12}$$

Thus, the molar fraction of each gas (N2, O2, water vapor) in the exhaust gas from the cathodic circulation pump can be expressed as:

$$\begin{cases} n_{O_2,ccp} = x_{O_2,dryair,ccp}(1 - x_{v,ccp})\dfrac{W_{ccp}}{M_{ccp}} \\ n_{N_2,ccp} = (1 - x_{O_2,dryair,ccp})(1 - x_{v,ccp})\dfrac{W_{ccp}}{M_{ccp}} \\ n_{v,ccp} = (1 - x_{v,ccp})\dfrac{W_{ccp}}{M_{ccp}} \end{cases} \tag{13}$$

According to equation (10)–(13), the gas constant of the gas mixture can be calculated as:

$$M_{mix} = \frac{W_{sm,in}}{(n_{O_2,cp} + n_{N_2,cp} + n_{v,cp}) + (n_{O_2,ccp} + n_{N_2,cp} + n_{v,ccp})} \tag{14}$$

The intersection of the two gas streams inevitably produces a change in temperature. The law of conservation of energy is used to determine the temperature of the mixed gas T_{mix}. Since the mixing chamber is located at the outlet of the air compressor and the exhaust gas recirculation pump, where the temperature difference between the two is small and the humidity difference is large, the model assumes that no liquid water is generated here. In fact, since the cathode recirculation operates under low current conditions, the low water content in the cathode recirculation gas makes the mixed gas not condense. The enthalpy of the gas before and after mixing is constant as follows.

$$H_{mix} = H_{cp} + H_{ccp} \tag{15}$$

The enthalpy of the air from the compressor consists of two parts: the enthalpy of dry air and the enthalpy of water vapor, calculated as follows:

$$H_{cp} = W_{cp} h_{cp} = 1.006 W_{dryair,cp} T_{cp,out} + W_{v,cp}(2501 + 1.86 T_{cp,out}) \tag{16}$$

Similarly, the enthalpy of the gas from the circulation pump is calculated as follows:

$$H_{ccp} = W_{ccp} h_{ccp} = 1.006 W_{dryair,ccp} T_{ccp,out} + W_{v,ccp}(2501 + 1.86 T_{ccp,out}) \tag{17}$$

Ultimately, the enthalpy of the mixed gas is

Table 2 – Mathematical equations for manifold dynamics [34].

Physical quantity	Governing equation
Supply manifold inlet flow	$W_{sm,in} = W_{cp} + W_{ccp}$
Supply manifold outlet flow	$W_{sm,out} = W_{ca,in} = k_{sm,out}(p_{sm} - p_{ca})$
Supply manifold pressure	$\dfrac{dp_{sm}}{dt} = \dfrac{T_{mix} R_{mix}}{V_{sm}}(W_{sm,in} - W_{sm,out})$
Return manifold inlet flow	$W_{rm,in} = W_{ca,out} = k_{ca,out}(p_{ca} - p_{rm})$
Return manifold outlet flow	$W_{rm,\ out} = W_{bpv,\ out} + W_{ccp}$
	$W_{bpv,\ out} = \begin{cases} \dfrac{C_D \pi D^2 (1-\cos(\theta)) p_{rm}}{4\sqrt{RT_{st}}}\left(\dfrac{p_{atm}}{p_{rm}}\right)^{\frac{1}{\gamma}}\sqrt{\dfrac{2\gamma}{\gamma-1}\left[1-\left(\dfrac{p_{atm}}{p_{rm}}\right)^{\frac{\gamma-1}{\gamma}}\right]}, & \dfrac{p_{atm}}{p_{rm}} > \left(\dfrac{2}{\gamma+1}\right)^{\frac{\gamma}{\gamma-1}} \\ \dfrac{C_D \pi D^2 (1-\cos(\theta)) p_{rm}}{4\sqrt{RT_{st}}}\sqrt{\gamma\left(\dfrac{2}{\gamma+1}\right)^{\frac{\gamma+1}{(\gamma-1)}}}, & \dfrac{p_{atm}}{p_{mm}} \le \left(\dfrac{2}{\gamma+1}\right)^{\frac{\gamma}{\gamma-1}} \end{cases}$
Return manifold pressure	$\dfrac{dp_{rm}}{dt} = \dfrac{T_{st} R_{air}}{V_{rm}}(W_{ca,\ out} - W_{rm,\ out})$

$$H_{mix} = W_{mix} h_{mix} = 1.006 W_{dryair,mix} T_{mix} + W_{v,mix}(2501 + 1.86 T_{mix}) \tag{18}$$

Ideal intercooler model

Since the high-temperature gas from the air compressor and the cathode circulation pump may cause damage to the membrane, the intercooler is an essential component to cool the inlet gas temperature to the desired temperature. In this paper, the intercooler is an ideal cooling model that fixes the gas temperature at 338 K, which is consistent with the reactor operating temperature. The humidity of the inlet gas passing through the intercooler can be obtained from the following equation [35]:

$$RH_{cl} = \frac{p_{v,cl}}{p_{sat}(T_{st})} = \frac{RH_{mix} p_{sat}(T_{mix})}{p_{sat}(T_{st})} \tag{19}$$

Where, the saturated vapor pressure $P_{sat}(T_{st})$ at the cooling temperature T_{st} can be expressed by the following equation:

$$\log_{10}(p_{sat}) = -1.69 \times 10^{-10} T^4 + 3.85 \times 10^{-7} T^3 - 3.39 \times 10^{-4} T^2 + 0.143 T - 20.92 \tag{20}$$

PEM fuel cell model

Voltage model

The voltage model of a fuel cell is a semi-mechanical and semi-empirical model. The output voltage of the fuel cell is obtained based on the energy-strength open-circuit voltage, activation loss, ohmic loss and concentration loss is determined as a closely related function of current, stack temperature, hydrogen-oxygen partial pressure and membrane humidity. The output voltage equation can be expressed as [36]:

$$v_{cell} = E_{Nernst} - v_{act} - v_{ohm} - v_{conc} \tag{21}$$

The Nernst open circuit voltage represents the thermodynamic ideal voltage of the fuel cell and can be expressed by the following equation:

Table 3 – The identified parameters and intrinsic parameters of the PEMFC model.

Parameters	Description	Values
N	Cell number	360
A_{cell}	Fuel cell active area	300 cm^2
i_{max}	Limiting current density	2.2 A/cm^2
F	Faraday constant	96,485 C/mol
c	empirical constant	0.02 [35]
k_1	identified coefficient	0.1406
k_2		-8.4×10^{-3}
k_3		-5.01×10^{-4}
		4.31×10^{-5}
k_5		-1.34×10^{-2}
k_6		0.2472
k_7		0.5746

$$E_{Nernst} = 1.229 - 8.5 \times 10^{-4}(T_{st} - 298.15) + \frac{RT_{st}}{2F}\left[\ln\left(p_{H_2}\right) + \frac{1}{2}\ln\left(p_{O_2}\right)\right] \tag{22}$$

The activation loss can be obtained from the following empirical equation:

$$v_{act} = k_1 + k_2 T_{st} + k_3 T_{st} \ln(C_{O_2}) + k_4 T_{st} \ln(I) \tag{23}$$

Further, the relationship between the dissolved oxygen concentration C_{O_2} at the cathode reaction interface and the oxygen partial pressure p_{O_2} can be obtained from Henry's law as follows.

$$C_{O_2} = 1.969 \times 10^{-7} p_{O_2} \exp(-T_{st}/498) \tag{24}$$

Ohmic loss includes contact resistance and proton resistance and can be expressed as follows:

$$\begin{cases} v_{ohm} = i \times R_{ohm} \\ R_{ohm} = \dfrac{l_m}{(k_5 \lambda_m + k_6)\exp\left(k_7\left(\frac{1}{303} - \frac{1}{T_{st}}\right)\right)} \end{cases} \tag{25}$$

The concentration loss due to mass transport can be determined by:

$$v_{conc} = c\ln\left(\frac{i_{max}}{i_{max} - i}\right) \tag{26}$$

According to Eqs. (22)–(26), the fuel cell terminal voltage equation (21) can be further expressed as:

$$v_{cell} = E_{Nernst} - \left(k_1 + k_2 T_{st} + k_3 T_{st} \ln\left(1.969 \times 10^{-7} p_{O_2} \exp(-T_{st}/498)\right) + k_4 T_{st} \ln(I)\right) - i \times \frac{l_m}{(k_5 \lambda_m + k_6)\exp\left(k_7\left(\frac{1}{303} - \frac{1}{T_{st}}\right)\right)} - c\ln\left(\frac{i_{max}}{i_{max} - i}\right) \tag{27}$$

In equation (27), $k_1 \sim k_7$ are the parameter coefficients and c is an empirical constant. Given the measured experimental data set, the parameters of the above voltage model are identified to obtain a higher fitting accuracy of the voltage model. In this paper, the values of the identified voltage model parameters are shown in Table 3.

Membrane hydration model

It is known from the voltage model that the output voltage is closely related to the average water content of the membrane λ_m. In general, the water content on the proton exchange membrane is mainly influenced by the electro-osmotic drag of protons and the reverse diffusion of water concentration across the membrane. The water flux across the membrane is determined by the following equation [35].

$$W_{v,mem} = M_{H_2O} A_{cell} N \left(n_d \frac{i}{F} - D_w \frac{c_{v,ca} - c_{v,an}}{l_m}\right) \tag{28}$$

Where, electro-osmotic drag coefficient n_d, water diffusion coefficient D_w as a function of the average water content of the membrane λ_m, water concentration $c_{v,ca}$ and $c_{v,an}$ as a function of the membrane water content of the corresponding side. The calculated equations for the above parameters are summarized in Table 4.

The unique variable λ_i (i is either ca-cathod, an-anode, m-membrane) is conservatively determined by the corresponding activity a_i as follow:

$$\lambda_i = \begin{cases} 0.043 + 17.81a_i - 39.85a_i^2 + 36.0a_i^3 & ,0 < a_i \le 1 \\ 14 + 1.4(a_i - 1) & ,1 < a_i \le 3 \end{cases} \tag{29}$$

Where, the anode activity a_{an} and the cathode activity a_{ca} are equal to the relative humidity of the corresponding side, the activity of the membrane a_m is expressed as the average of the cathode and anode activities.

Cathode and anode flow model

Since the output voltage of the fuel cell is closely related to the partial pressures of oxygen, hydrogen and water vapor on both sides of the cell, it is necessary to solve for the above states inside the cell. The mass continuity equation and the electrochemical reaction relationship determine the controlling equations for the partial pressures of oxygen, nitrogen, and water vapor on the cathode side as follows [38]:

$$\frac{dp_{O_2}}{dt} = \frac{R_{O_2} T_{st}}{V_{ca}} \left(W_{O_2,in} - W_{O_2,out} - W_{O_2,react} \right) \tag{30}$$

$$\frac{dp_{N_2}}{dt} = \frac{R_{N_2} T_{st}}{V_{ca}} \left(W_{N_2,in} - W_{N_2,out} \right) \tag{31}$$

$$\begin{cases} \frac{dp_{v,ca}}{dt} = \frac{R_{H_2O} T_{st}}{V_{ca}} \left(W_{v,in,ca} - W_{v,out,ca} + W_{v,gen} + W_{v,mem} \right) \text{if } p_{v,ca} < p_{sat}(T_{st}) \\ p_{v,ca} = p_{sat}(T_{st}) \text{ if } p_{v,ca} \ge p(T_{st}) \end{cases} \tag{32}$$

Table 4 – Calculation of membrane hydration parameters [35,37].

Parameters	Mathematical equations
n_d	$0.0029\lambda_m^2 + 0.05\lambda_m - 3.4 \times 10^{-19}$
D_w	$D_\lambda \exp\left(2416\left(\frac{1}{303} - \frac{1}{T_{st}}\right)\right)$
D_λ	$\begin{cases} 10^{-6} & ,\lambda_m < 2 \\ 10^{-6}(1 + 2(\lambda_m - 2)) & ,2 \le \lambda_m \le 3 \\ 10^{-6}(3 - 1.67(\lambda_m - 3)) & ,3 < \lambda_m < 4.5 \\ 1.25 \times 10^{-6} & ,\lambda_m \ge 4.5 \end{cases}$
$c_{v,an}$	$\frac{\rho_{m,dry}}{M_{m,dry}}\lambda_{an}$
$c_{v,ca}$	$\frac{\rho_{m,dry}}{M_{m,dry}}\lambda_{ca}$

Table 5 – Structural parameters and general constants for PEMFC system model.

Parameters	Description	Values	References
V_{sm}	Supply manifold volume	0.02 m^3	[35]
V_{rm}	Return manifold volume	0.005 m^3	[35]
V_{ca}	Cathode volume	0.01 m^3	[35]
V_{an}	Anode volume	0.005 m^3	[35]
$k_{sm,out}$	Cathode inlet flow constant	2.2×10^{-4} kg/(s*Pa)	[34]
$k_{ca,out}$	Cathode outlet flow constant	3.6×10^{-4} kg/(s*Pa)	[34]
J_{cp}	Inertia of the compressor and the motor	1×10^{-4} kg*m^2	[35]
k_v	Motor constant	0.0153 V/(ras/s)	[39]
k_t	Motor constant	0.0225 Nm/A	[35]
R_{cm}	Motor constant	1.2 Ω	[35]
η_{cm}	Compressor motor efficiency	0.8	[35]
C_p	Specific heat capacity of air	1.004 J/kg*K	[40]
γ	Ratio of specific heat of air	1.4	[40]
C_D	Discharge coefficient of nozzle	0.012	[40]
R	Universal gas constant	8.3415 J/(mol*K)	[41]
R_{O_2}	Oxygen gas constant	259.8 J/(kg*K)	[41]
R_{N_2}	Nitrogen gas constant	296.8 J/(kg*K)	[41]
R_{H_2O}	Vapor gas constant	461.5 J/(kg*K)	[41]
M_{O_2}	Oxygen molar mass	32 g/mol	[40]
M_{N_2}	Nitrogen molar mass	28 g/mol	[40]
M_{H_2O}	Vapor molar mass	18 g/mol	[40]
P_{atm}	Atmosphere pressure	1.103 atm	[35]
M_{H_2}	Hydrogen molar mass	2 g/mol	[35]
R_{H_2}	Hydrogen gas constant	4.124 J/(kg*K)	[35]
T_{atm}	Atmosphere temperature	298 K	[34]

$W_{O_2,react}$ and $W_{v,gen}$ are the oxygen consumption and water production of the electrochemical reaction, respectively, calculated as follows

$$W_{O_2,react} = \frac{NI}{4F} \times M_{O_2} \tag{33}$$

$$W_{v,gen} = \frac{NI}{2F} \times M_{H_2O} \tag{34}$$

Similarly, the hydrogen partial pressure and water vapor partial pressure on the anode side are given by the mass conservation equation:

$$\frac{dp_{H_2}}{dt} = \frac{R_{H_2} T_{st}}{V_{an}} \left(W_{H_2,in} - W_{H_2,out} - W_{H_2,react} \right) \tag{35}$$

$$\begin{cases} \dfrac{dp_{v,an}}{dt} = \dfrac{R_{H_2O} T_{st}}{V_{an}} \left(W_{v,in,an} - W_{v,out,an} - W_{v,mem} \right) \; if \; p_{v,an} < P_{sat}(T_{st}) \\ p_{v,ca} = P_{sat}(T_{st}) \; if \; p_{v,an} \geq P_{sat}(T_{st}) \end{cases} \tag{36}$$

According to Faraday's law, the mass flow of hydrogen involved in the reaction at the anode is:

$$W_{H_2,react} = \frac{NI}{2F} \times M_{H_2} \tag{37}$$

Since the hydrogen supply manifold is small, its volume is lumped together with the anode volume in the model, i.e. they have the same pressure. The flow rate of the anode inlet is determined using a proportional controller where the anode pressure follows the cathode pressure:

$$W_{an,in} = \xi_1 (P_{ca} - P_{an}) \tag{38}$$

A control model of an automotive fuel cell system with cathode recirculation that simulates the real operating behavior can be built by using the modeling equations presented in Sections Air compressor-PEM fuel cell model. The structural parameters and generic constants used in the system model are given in Table 5.

Model validation

In sections System structure and model hypothesis and Air compressor, the developed compressor and cathode circulation pump models are experimentally verified and high precision relationships between flow, pressure and speed are obtained, which provide the basis for the development of the system model. In order for the model to obtain output performance close to that of the actual bench, it is necessary to compare the developed stack model with the actual experimental data. The experiments are conducted on the vehicle fuel cell system bench shown in Fig. 1. Since this paper is more concerned with the problem of high potential under small current density conditions, more validation work is carried out under low load conditions.

The steady-state performance of the stack model is verified by polarization curves as shown in Fig. 4(a). Since this study focuses on the low load or idling conditions of the vehicle, the steady-state voltages at different current points in the range of 0–120 A are extensively verified. It can be seen that the voltage and power predicted by the model are highly close to the real values obtained by the experiment; moreover, Fig. 4(d) further gives the relative errors between them, which are almost below 1% for the whole operating conditions. Therefore, from these above steady-state tests, it is evident that the

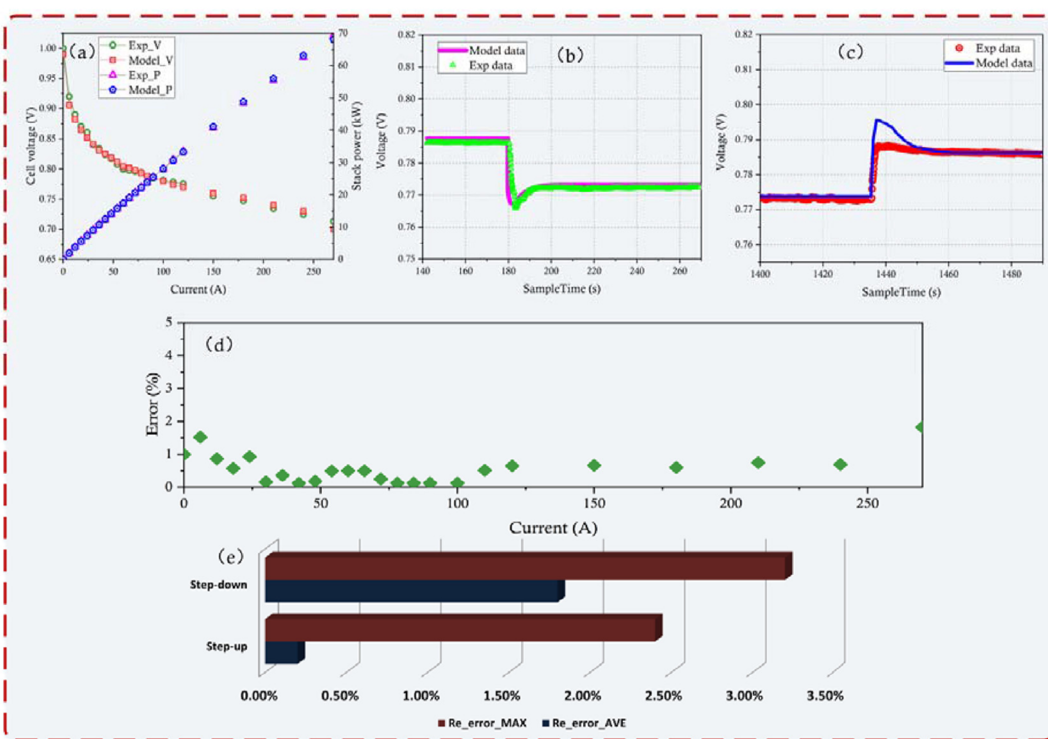


Fig. 4 – Comparative analysis of experimental and simulation results of output performance.

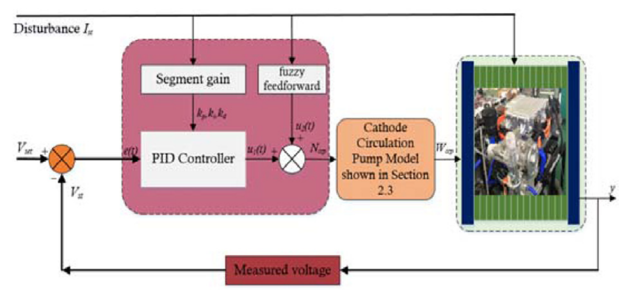


Fig. 5 – Structure of feedforward compensation + segmented PID controller.

developed model can characterize the steady-state performance of the actual fuel cell system with high accuracy. Fig. 4(b) and (c) verify the voltage transient behavior of the model in the step-up and step-down stages, respectively. The results show that the model is able to qualitatively simulate the transient behavior of the actual system. From the error analysis in Fig. 4(e), the maximum relative error and average relative errors are 2.3% and 0.2% in the step-up stage, and 3.2% and 1.8% in the step-down stage, respectively. From the steady-state and transient verification results, the model exhibits satisfactory simulation accuracy. This permits to evaluate the new control design as closely as a real system and to check the evolution of many internal variables.

High-potential controller design

The high potential problem associated with the long-term operation of automotive fuel cell systems at low load or idle conditions is a great challenge for durability enhancement. Therefore, the rapid and accurate control of high potentials under system dynamics through cathode recycling makes fuel cells operate at relatively low voltages, which has positive significance for mitigating carbon corrosion and improving durability. In this paper, a hybrid control scheme with dynamic fuzzy logic PID with feedforward compensation is designed to achieve accurate and fast control of high potentials based on the established model of a vehicle fuel cell system with cathode recirculation, and it is compared with a single control strategy.

Segmented PID controller

Conventional PID control is a feedback control method widely configured in industrial process control with the advantages of simplicity of structure, ease of deployment and high control efficiency for both linear and nonlinear systems. The expression for the time domain function can be expressed as [40]:

$$u(t) = k_p e(t) + k_i \int_0^t e(t)dt + k_d \frac{de(t)}{dt} \tag{39}$$

Where, $e(t)$ is the feedback error, $u(t)$ is the output of the controlled variable, and k_p, k_i, k_d are the proportional, differential, and integral gains, respectively. However, in the operation of a real PEMFC system, the complexity and variability of the operating conditions make it impossible to always obtain the best control effect with a single gain, e.g., the inevitable steady-state error. The empirical segmentation gain at different operating points solves the above problem to some extent. The segmented PID controller can further improve the robustness of control by obtaining different regulation gains through calibration experiments.

Segmented PID controller with feedforward

In a single segmented PID feedback control, the feedback is controlled by deviation. When disturbances are constantly present, the system first generates deviations, and then the controller generates control actions to offset the effects of the disturbances in accordance with the deviations. This indicates that the system always starts to change after the disturbance, and it is difficult for the controller to overcome the system deviation caused by the disturbance in time. Therefore, a single segmented PID controller has the following drawbacks: (i) slow response time when the disturbance comes to system stability; ii) high regulation pressure for frequent changes of different disturbances. Feedforward compensation control solves the above problems caused by closed-loop negative feedback regulation to some extent. The feedforward control measures the disturbance variables entering the process (including external disturbances and setpoint changes) and generates appropriate control actions based on the disturbance measurements so that the control variables can be quickly maintained near the setpoint. Therefore, the combination of feedforward and segmental PID can further improve the response speed and control accuracy. The structure diagram of feedforward compensation + segmental PID controller acting on the controlled system is shown in Fig. 5. Feedforward compensation + segmented PID controller is better than single PID control in reducing response time. However, as a static control method, it is still inadequate to solve the issues of overshoot and responsiveness caused by dynamic disturbances. In the actual automotive fuel cell system, the controlled object is strongly nonlinear and time-varying. Further development of dynamic adaptive control strategies is needed to ensure control quality.

Dynamic fuzzy logic PID controller with feedforward compensation

Dynamic adaptive control is particularly important due to the random changes in the fuel cell system. Fuzzy logic control as an intelligent control method does not depend on high-precision mathematical models in practical applications and is particularly suitable for complex fuel cell systems with nonlinear variations [42,43]. Also, relying on the established fuzzy rule base, fuzzy logic control has superior adaptability and robustness [44,45]. Combining feedforward segmented PID with fuzzy logic algorithm for fuel cell nonlinear system control is a promising solution. In the framework of this combined strategy, the PID algorithm is the main body and the fuzzy logic controller calculates the weighting factors through the designed rule base and adaptively adjusts the gain parameters to achieve better control results. The structure of the proposed combinatorial strategy of dynamic fuzzy logic PID

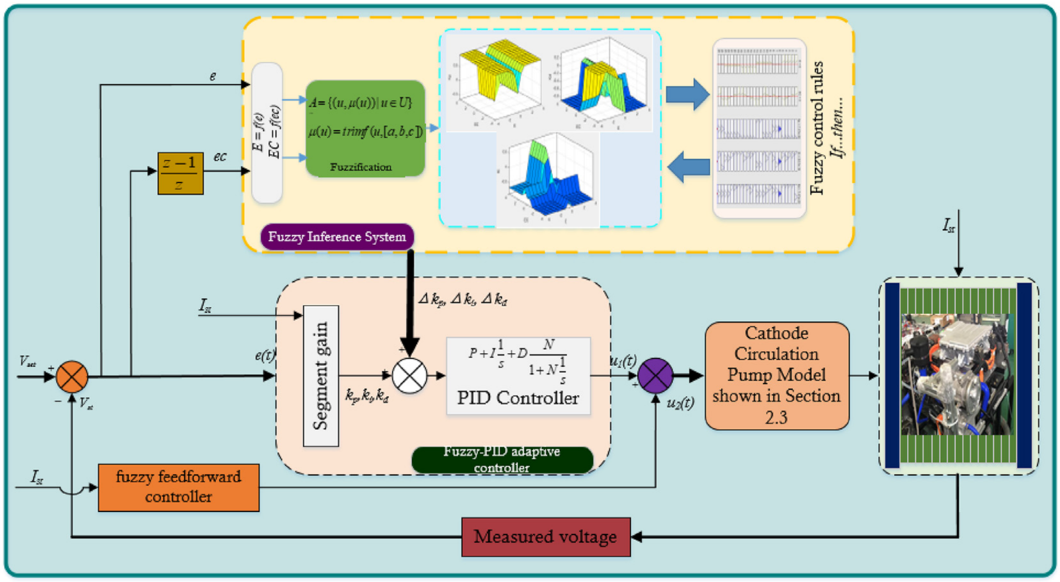


Fig. 6 – Controller structure of the proposed dynamic fuzzy logic PID with feedforward compensation.

with feedforward compensation in this paper is shown in Fig. 6.

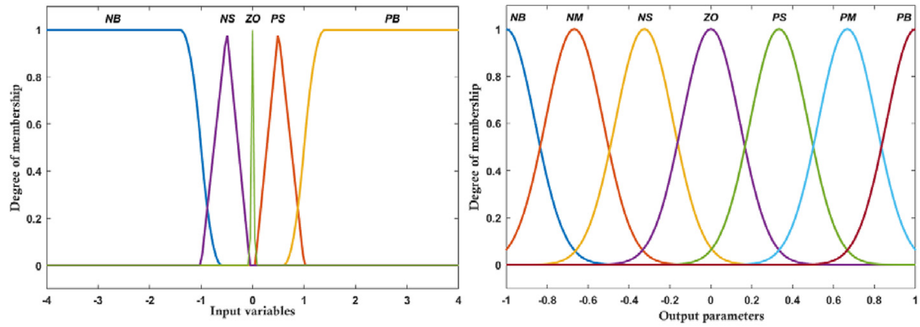
The key to achieving adaptability in a PEMFC system is the design of a fuzzy logic controller [46]. The design process of a fuzzy logic controller consists of three important steps. The first step is fuzzification, which includes the transformation of the domain of the input variables and the fuzzification process; the second step is to design fuzzy rules and perform fuzzy inference to obtain the preliminary output; the last step is to clarify the fuzzy results and convert them to the actual control output [47]. In this paper, a two-dimensional fuzzy logic controller is designed for adaptive regulation of gains, and the specific design process is as follows:

Fuzzy logic controller design

Through extensive simulations of fuel cell systems without fuzzy control strategies, the actual theoretical domain with respect to the control error (e) and the error difference (ec) is determined and used as the initial input variables for the fuzzy logic controller. The actual domains of the input variables are transformed into fuzzy set domains by means of mapping functions. In this paper, the fuzzy subset theoretical domain corresponding to the error and error difference of the voltage is E&EC belongs to −4 to 4, which was obtained after extensive analysis of simulation results. The output of the fuzzy controller is the three gains of the PID controller, and they are normalized in the controller to any real number in the range −1 to 1 and multiplied by the corresponding coefficient matrix to obtain the final output value. After obtaining the fuzzy domain, the design of fuzzy subsets of the input and output variables is carried out. Fuzzy subsets play an important bridging role as a trade-off between control effects and fuzzy rules. Generally, the seven-level quantification standard is generally adopted for the formulation of fuzzy subsets, but this also depends on the specific control system. Considering the operation of the actual system in this paper and the fast real-time nature of the control, the fuzzy subsets of the input variables are adopted with five levels of quantization: NB (much less than 0), NS (slightly less than 0), ZO (equivalent to 0), PS (slightly greater than 0), and PB (much greater than 0). The fuzzy subsets of output variables are quantified using seven levels: NB, NM (moderately less than 0), NS, ZO, PS, PM (moderately greater than 0), PB. The membership function is the mapping relationship between fuzzy variables and fuzzy subsets, which is related to the sensitivity and precision of control. In this paper, a specific input-output membership function is designed based on the subjective knowledge and experience of the simulation system. The membership function of the input variables combines the characteristics of Z-type, S-type and triangular functions, and its specific schematic is shown in Fig. 7(a). The membership function of the output variable considers a more stable Gaussian-type function, as shown in Fig. 7(b). The mathematical expressions of the above membership functions are shown in Eqs. (38)–(41).

$$f(x;a,b)=\begin{cases}1, & x\le a\\ 1-2\left(\dfrac{x-a}{b-a}\right)^2, & a\le x\le \dfrac{a+b}{2}\\ 2\left(\dfrac{x-b}{b-a}\right)^2, & \dfrac{a+b}{2}\le x\le b\\ 0, & x\gg b\end{cases} \tag{40}$$

$$f(x;a,b,c)=\begin{cases}0, & x\le a\\ \dfrac{x-a}{b-a}, & a\le x\le b\\ \dfrac{c-x}{c-b}, & b\le x\le c\\ 0, & x\gg c\end{cases} \tag{41}$$



(a) (b)

Fig. 7 – The designed membership function. (a) For input variables E&EC, (b) for output parameters k_p, k_i, k_d.

$$f(x;a,b)=\begin{cases}0\,, & x\le a\\ 2\left(\dfrac{x-b}{b-a}\right)^2, & a\le x\le \dfrac{a+b}{2}\\ 1-2\left(\dfrac{x-a}{b-a}\right)^2, & \dfrac{a+b}{2}\le x\le b\\ 1\,, & x\gg b\end{cases} \tag{42}$$

$$f(x;\sigma,c)=e^{\frac{-(x-\bar{x})^2}{2\sigma^2}} \tag{43}$$

Where, Eq (40)–(43) are expressions for Z-type, triangular-type, S-type and Gaussian-type functions, respectively. x is the input variable, a,b,c are the parameters specified in the function, σ is the standard deviation, and $\bar{x}$ is the mean value.

The establishment of fuzzy rules is the most important part of fuzzy controller design, and the design criterion is to ensure that the control result has a smaller steady-state error, faster response time and smaller overshoot. The fuzzy control rules designed in this paper are established based on the working experience obtained from a large number of system simulation experiments. Firstly, the operation laws of the fuel cell system with cathode recirculation are analyzed: The PEMFC system consists of two gas supply units, the compressor and the circulation pump. To reduce the stress on the control system, the compressor speed is regulated by a linear control method to maintain the fresh air flow at the required oxygen excess ratio (OER). After the recirculation pump is turned on, the balance of the system is disturbed. The most immediate effect is that the increase in supply manifold pressure causes the compressor to deviate from its previous operating condition. This can be summarized as an increase in circulation pump flow inevitably leads to a lower compressor outlet flow due to supply interference problems. This operational interference positively contributes to the reduction of the proportion of oxygen in the intake air. The principle of the fuzzy rule is that when the actual voltage is higher than the target voltage, the speed of the circulating pump should be increased while limiting the surge caused by the pressure rise of the air compressor; when the actual voltage is lower than the target voltage, the speed of the circulating pump should be reduced. According to the above principles, a fuzzy rule base for the full range of simulation conditions is obtained as shown in Table 6. Finally, a fuzzy output is obtained by the Mamadani inference system based on the designed fuzzy rule base.

The fuzzy output is defuzzified using the centroid method, which uses the median in the fuzzy control system as the control variable and contains all the information in the fuzzy subset. The centroid method is calculated as follows.

$$\Delta k_{p,i,d}=\frac{\sum_{i=0}^{n}\mu_i(z_i)*z_i}{\sum_{i=0}^{n}\mu_i(z_i)} \tag{44}$$

Where, $\Delta k_{p,i,d}$ is the exact value of the fuzzy output value after defuzzification, which in this case represents the regulation of the PID gain, z_i is the fuzzy quantization value, and $\mu_i(z_i)$ is the membership degree of z_i.

With a well-designed fuzzy logic controller, a series of gain adjustments $\Delta k_{p,i,d}$ can be obtained at different operation stages. Thus, the final gain of the PID controller can be calculated by the following equation.

$$k_{p,i,d_{final}}(i)=\Delta k_{p,i,d}(i)+k_{p,i,d}(i) \tag{45}$$

Where, $k_{p,i,d\_final}(i)$ is the final gain under different operating conditions, $\Delta k_{p,i,d}(i)$ is the gain adjustment amount of the

Table 6 – The designed fuzzy rule base.

U_0		EC				
		NB	NS	ZO	PS	PB
E	NB	PB NB PS	PB NM NB	PB NM NM	PB NS NB	PB NM NB
	NS	PM NB ZO	PM NS NS	PM NB NM	NB PS ZO	NB PS ZO
	ZO	ZO NM ZO	ZO NS NS	ZO ZO ZO	ZO PS NS	ZO PM ZO
	PS	PM NB NB	PM NB NM	PM NM NM	PM NS ZO	PM NS ZO
	PB	PB NB NB	PB NS NM	PB NM NM	PB NM NS	PB NB NB

fuzzy controller output, and $k_{p,i,d}(i)$ is the original gain of the PID controller.

Overall, the designed dynamic fuzzy logic controller with feedforward compensation has the following advantages: feedforward ensures fast response to overcome random disturbances, PID feedback control ensures the elimination of steady-state errors, and fuzzy logic achieves dynamic control of following disturbances in terms of overshoot and responsiveness.

Results and discussion

In this section, using the fuel cell system model developed in Section Modeling and validation of the PEMFC system with cathode recirculation, several different high-potential controllers introduced in Section High-potential controller design are developed and evaluated, which have positive implications for the mitigation of durability degradation caused by high potentials when the automotive fuel cell system is operating at low load or idle conditions. Meanwhile, the impact of important states and variables, especially humidity, caused by the installation of a cathode recirculation pump are analyzed.

Evolution of state variables

The transient evolution process of the important internal states and variables of the fuel cell system before and after the addition of cathode recirculation is evaluated as shown in Fig. 8. The implemented current profile is shown in Fig. 8(a) with the fuel cell system operating at low load conditions loaded from 30 A to 100 A with a total operating time of 50 s. Fig. 8(i) gives the cathode recirculation ratio basically set in the range of 45%–50% for the whole operating conditions. The cathode recirculation ratio is determined as the flow rate at the outlet of the recirculation pump divided by the flow rate at the outlet of the stack cathode. The rotational speed of the cathode circulation pump is shown in Fig. 8(g), which is positively correlated with the current profile. It can be seen from Fig. 8(b) that the addition of cathode recirculation increases the supply pressure P_{sm} accordingly. Furthermore, Fig. 8(c) and (d) shows the effect of cathode recirculation on the oxygen partial pressure and nitrogen partial pressure inside the PEMFC. Compared to the normal mode without cathode recirculation, the cathode recirculation mode increases the nitrogen partial pressure and decreases the oxygen partial pressure because the recirculated cathode gas becomes a gas with an increased nitrogen-to-oxygen ratio due to the oxygen consumption behavior of the electrochemical reaction. When mixed with the compressor flow, the mixed gas pressure increases as shown in Fig. 8(b) and the nitrogen-to-oxygen ratio also increase. According to Dalton's law of partial pressure, it can be obtained that the nitrogen partial pressure increases while the oxygen partial pressure decreases after adding the cathode recirculation. Another important change is the return manifold pressure P_{rm}. The cathode recirculation mode causes the pressure in the return manifold to decrease because the addition of the recirculation pump increases the flow of the return manifold outlet and thus decreases the flow in the return manifold. The pressure is thus reduced according to the ideal gas law. The variation of the oxygen excess ratio is shown in Fig. 8(b), where the oxygen content of the mixed gas increases due to the addition of oxygen in the cathode recirculation gas.

Robustness analysis

Since the fuel cell system may experience unknown parameter changes during operation such as runaway compressor,

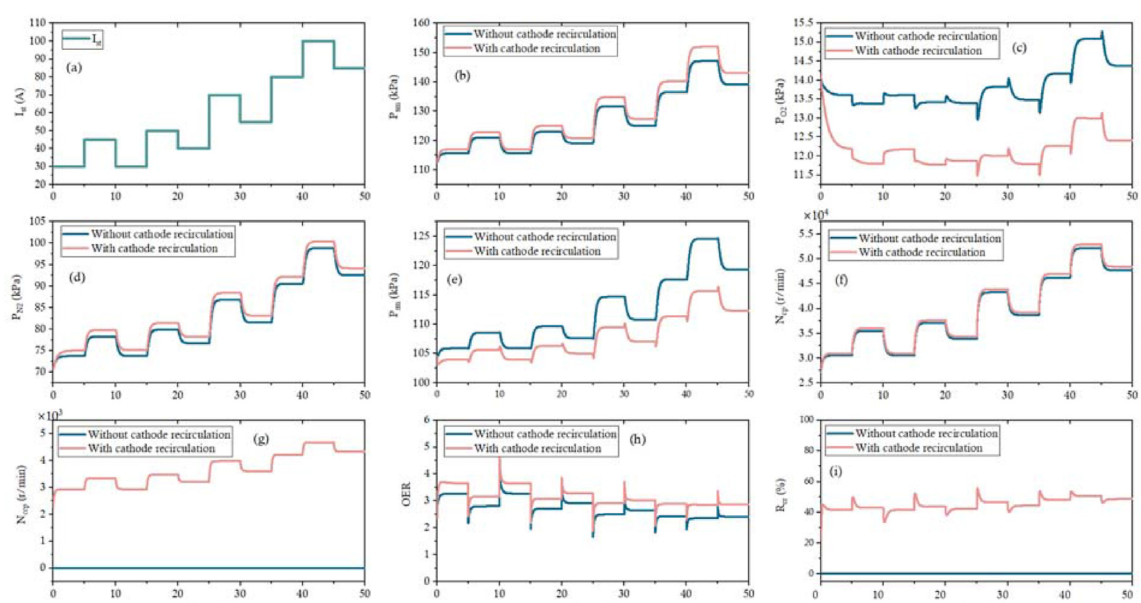


Fig. 8 – Evolution of key state variables in PEMFC system (without cathode recirculation *vs.* with cathode recirculation). (a) Current profile; (b) supply manifold pressure; (c) Partial pressure of oxygen; (d) Nitrogen partial pressure; (e) Return manifold pressure; (f) Air compressor speed; (g) Cathode circulating pump speed; (h) Oxygen excess ratio; (i) Cathode recycling ratio.

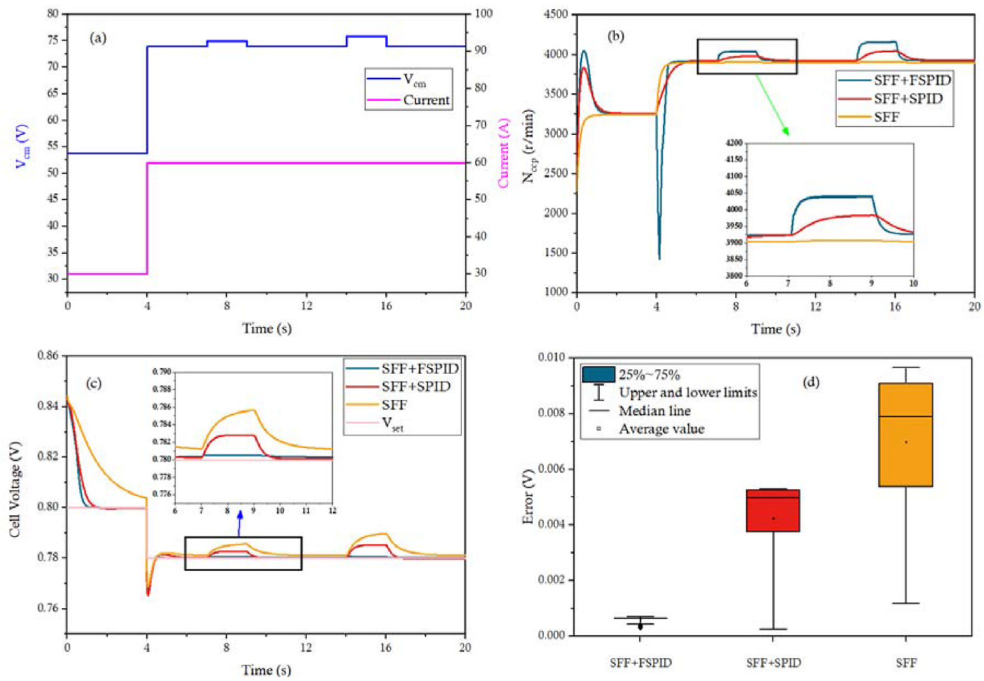


Fig. 9 – Robustness comparison of different control strategies. (a) Air compressor motor voltage; (b) Cathode circulating pump speed; (c) Cell voltage; (d) Voltage control error.

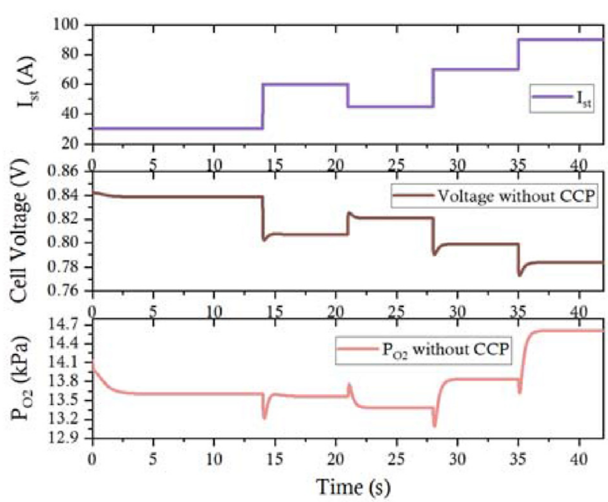


Fig. 10 – Current, voltage and oxygen partial pressure of PEMFC system without cathode recirculation.

jittering backpressure valve or blocked transmission channel. Therefore, it is required that the developed controller should have strong robustness against random disturbances. In this section, the robustness of the designed high-potential controller is compared and analyzed by setting the motor voltage of the air compressor as an uncertainty factor. This is because changing the compressor motor voltage changes the flow rate from the compressor, which further changes the oxygen partial pressure inside the cell, thus directly affecting the output voltage. Therefore, this is a strongly relevant uncertainty factor for achieving high potential control based on oxygen partial pressure regulation using cathode recirculation. The current profile for evaluating the robustness is shown in Fig. 9(a), rising from 30 A to 60 A. The current profile for evaluating robustness is shown in Fig. 9(a), loaded from 30 A to 60 A. Meanwhile, the air compressor motor voltage V_{cm} is the variable with uncertainty. The total running time is 20s, with disturbances occurring at 7s and 14s for a duration of 2s. Fig. 9(b) shows the speed regulation behavior of the cathode circulation pump produced by different controllers after the random disturbance is generated. The static feedforward control does not have anti-disturbance ability because it is an open-loop control, while the SFF+FSPID controller shows superior robustness. Specifically, when there is a sudden increase in the compressor voltage, the compressor outlet flow rate increases, and in order to reduce the oxygen partial pressure at the stack inlet, the cathode circulation pump needs to increase the speed in time to suppress the compressor flow rate and the total inlet oxygen content. It can be seen that the fast response of SFF+FSPID controller makes N_{ccp} rise from 3900r/min to 4050r/min. Fig. 9(c) shows the ability of the different controllers to adjust the output voltage after the disturbance occurs. The SFF+FSPID controller shows better robustness both in terms of regulation time and regulation accuracy. Despite the presence of unknown disturbances, the SFF+FSPID controller has almost no voltage

Table 7 – High potential limiting targets for low load conditions.

I_{st}/A	Normal voltage/V	Voltage control targets/V
30	0.84	0.8
45	0.82	0.8
60	0.81	0.78
70	0.795	0.78
90	0.78	0.76

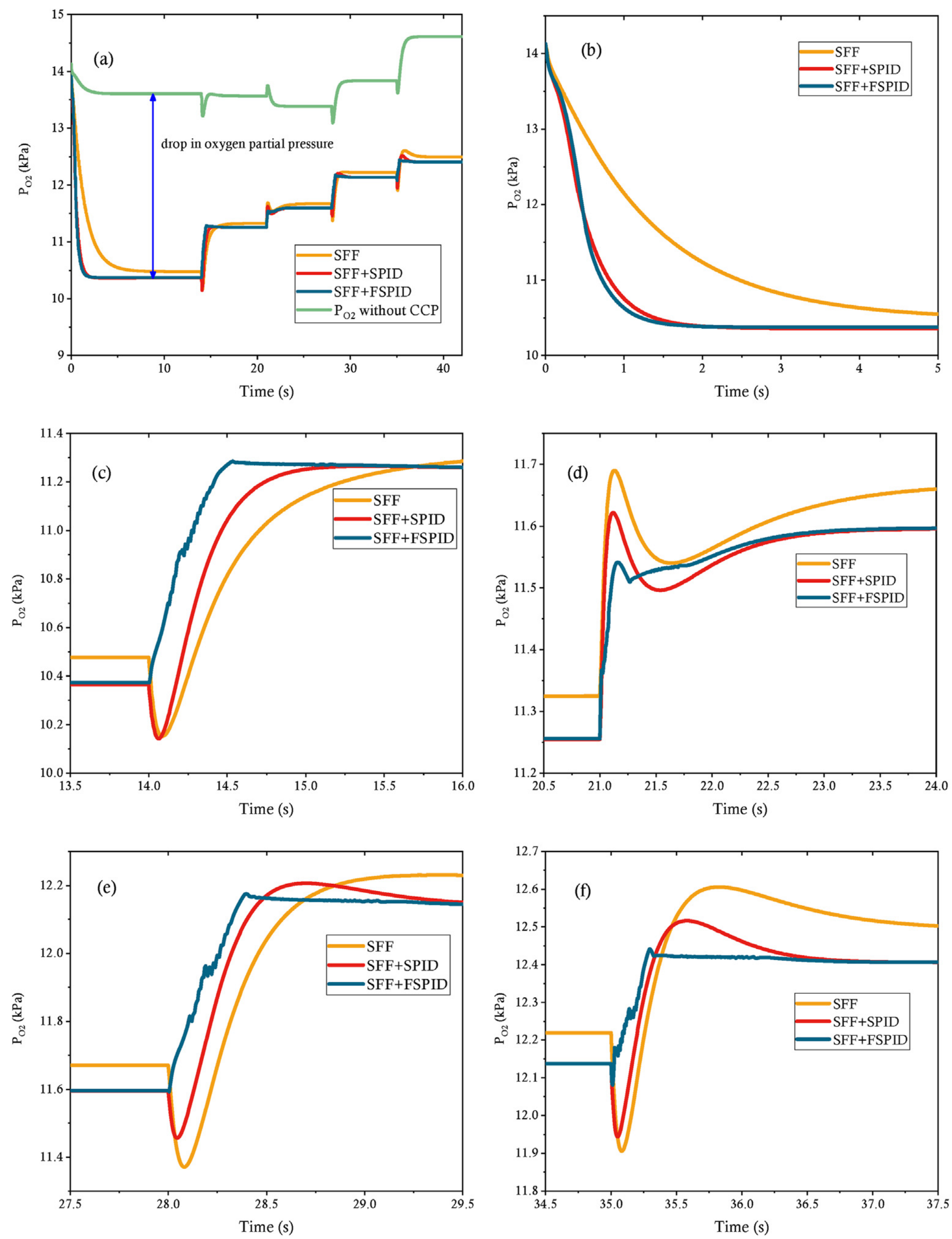


Fig. 11 – The effect of different high potential controllers on the partial pressure of oxygen at the cathode. (a) The whole process of oxygen partial pressure control effect; (b) Magnified view at start-up stage of controller; (c) Magnified view at 14s; (d) Magnified view at 21s; (e) Magnified view at 28s; (f) Magnified view at 35s.

deviation. Fig. 9(d) further analyzes the statistical control error during the duration of the disturbance. The average value of voltage error of SFF+FSPID is within 1 mV, which strictly tracks the target value.

Dynamic performance of oxygen partial pressure control

Fig. 10 presents the current profile of the system and shows the results of the voltage and internal oxygen partial pressure of the fuel cell without cathode recirculation, which will provide a control for the implementation of the high potential controller and the analysis of the results. The current operating curve is set to cross-vary from 30A to 90A at low load operating conditions. Table 7 shows the actual and target voltages at different currents under low load conditions. It can be seen that if no limiting measures are taken, the fuel cell will be exposed to high potential cycles above 0.8v for a long time, which will cause irreversible degradation of durability. Fig. 11 shows the oxygen partial pressure dynamics demonstrated by different high-potential controllers for the fuel cell system running under the current profile shown in Fig. 10 with Table 7 as the control target. Fig. 11(a) shows an overall control result compared to that without cathode recirculation. It can be seen that all three controllers are able to reduce the oxygen partial

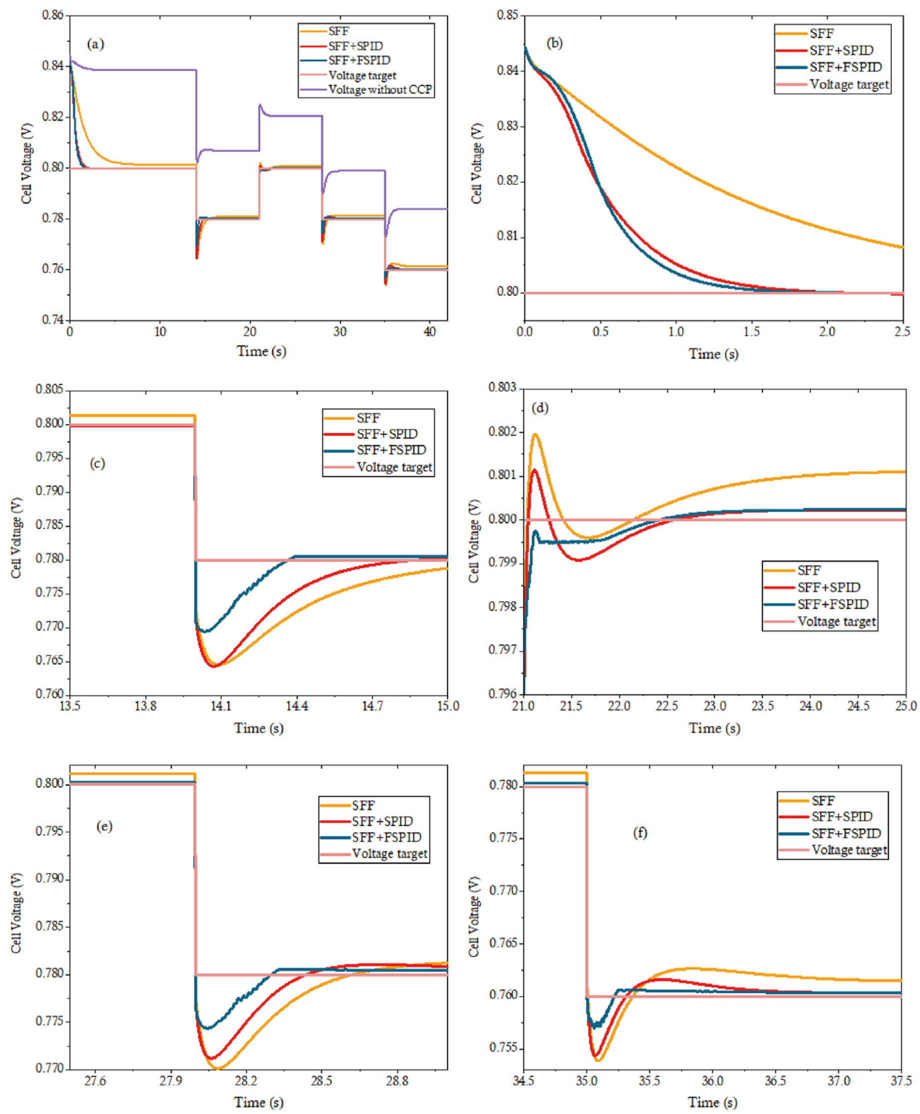


Fig. 12 – High potential nonlinear control performances of different controllers. (a) The whole process of cell voltage control effect; (b) Magnified view at start-up stage of controller; (c)–(f) Magnified view at 14s, 21, 28s and 35s.

pressure inside the fuel cell under low load conditions as much as possible, but the detailed control effect is not the same. Fig. 11(b)–(f) further show the magnified view of the oxygen partial pressure change at the moment of load change. Firstly, as shown in Fig. 11(b), the response time of the SFF controller lags far behind the other two controllers and has non-eliminable steady-state errors, however, the SFF + FSPID controller demonstrates superior response performance and control accuracy. Fig. 11(c) shows the effect of oxygen partial pressure regulation at the instant of current loading. Generally, in normal mode, the oxygen partial pressure of the PEMFC will drop and then rise at the moment of loading because the electrochemical reaction consumes oxygen much faster than the oxygen supply response, and the cathode recirculation will further aggravate this phenomenon, which is extremely detrimental to the stable voltage transition. It can be seen that both SFF and SFF + SPID controllers show the above change process. On the contrary, the SFF + FSPID controller eliminates the sudden drop of oxygen partial pressure through strong adaptive ability and significantly improves the dynamic response speed. Meanwhile, the same effect is observed during the loading in Fig. 11(e) and (f). Fig. 11(d) shows the control effect of the oxygen partial pressure in step-down stage. When the load is reduced, due to the excess of oxygen, a process in which the oxygen partial pressure first rises and then falls and finally stabilizes is observed. The SFF + FSPID controller obtains the smallest oxygen partial pressure overshoot, mitigates oscillations and follows the target value in the shortest possible time. It can be observed from Fig. 11(e)–(f) that when the loading is completed, SFF controller has a large steady-state error at higher currents, and SFF + SPID controller shows the traditional static PID controller common problem with unavoidable overshoot, but can achieve the elimination of steady-state error, the SFF + FSPID controller shows the most superior results, i.e., eliminating overshoot, further reducing steady-state errors and having the fastest response time. The analysis of the regulation effect of oxygen partial pressure when implementing different high-potential controllers can further provide a theoretical basis for voltage control.

Dynamic performance of high potential control

The high potential dynamic control performance under the action of different controllers is obtained as shown in Fig. 12 by implementing the current profile shown in Fig. 10 on the fuel cell system. In the fuel cell voltage model Equation (27), the voltage is considered as a function of current, oxygen partial pressure, temperature, membrane humidity, and hydrogen partial pressure, and the largest difference before and after the cathode recirculation configuration is the oxygen partial pressure. It can be concluded that when the system is equipped with cathode recirculation, the change in oxygen partial pressure determines the output voltage. Therefore, the change process of the voltage in Fig. 12 can be explained from the content revealed in Fig. 11. Fig. 12(a) shows an overall voltage dynamic following process, and it can be seen that all three controllers have voltage suppression capability, but there are differences during dynamic load change moments and steady-state following stage as shown in the enlarged view in Fig. 12(b)–(f). The SFF + FSPID controller is far ahead of the single SFF and SFF + SPID controllers in terms of response speed, both during the controller start-up phase and during the load-up and load-down phases. In particular, from the results of the voltage dynamics without cathode recirculation shown in Fig. 12(a), the voltage shows a stable recovery phenomenon during the period of stabilization after the variable load. However, this phenomenon is broken after configuring the cathode recirculation, which can be observed in Fig. 12(d) and (f) (21.3s–21.9s & 35.3s–36.5s). Since the cathode recirculation interferes with the change process of the intake pressure, the oxygen partial pressure in the corresponding stage as shown in Fig. 11(d)h and (f) changes, which makes the voltage fluctuate in the stable stage. From the control effects of the three controllers, the designed SFF + FSPID controller robustly eliminates the voltage fluctuations. In terms of steady-state error, the SFF controller performs the worst especially under high current conditions, and the SFF + FSPID controller shows the optimal steady-state following performance.

The quantitative comparison of the dynamic control performance exhibited by the different controllers is summarized in Table 8. The controller performance at the load-up and load-down moments represented in Fig. 12(d) and (f) are quantified in detail. Deviation (a) is defined as the voltage undershoots or overshoots caused by oxygen starvation or oxygen excess at the moment of load change. Deviation (a) has a different influence process in PEMFC systems with and without cathode recirculation. In a normal mode fuel cell system without cathode recirculation, voltage deviation occurs at the moment of load change because the response of the compressor and piping lags behind the electrochemical reaction. When cathode recirculation is introduced, a different phenomenon occurs when the system experiences a variable load. Specifically, at load-down moment, the addition of cathode recirculation relatively lowers the oxygen partial pressure in the inlet gas thereby alleviating voltage overshoot, but the response of cathode recirculation lags far behind the electrochemical reaction, so overshoot is still inevitable. At the load-up moment, the addition of cathode recirculation accelerates the drop in oxygen partial pressure in the intake air and thus increases the magnitude of the voltage undershoot. The deviation (b) is defined as the voltage deviation due to the inertia of the compressor and recirculation pump after voltage recovery. This voltage deviation is almost 0 in a normal system, as shown in the purple line in Fig. 12(a). However, the addition of cathode recirculation changes the original inertia of the inlet air, which results in voltage deviations as shown in Fig. 12(d) and (f). Therefore, in a PEMFC system equipped with cathode recirculation, the ideal voltage response process during load variation is to eliminate deviation (a) and deviation (b) as much as possible while keeping the voltage stable at the target voltage using the designed high potential controller based on oxygen partial voltage regulation. The tuning time is defined as the duration of the voltage deviation from the deviated state to the steady-state during the variable load. From the results in Table 8, the SFF+FPID controller demonstrates satisfactory dynamic regulation, i.e., the fastest tuning time and the smallest voltage deviation, both in the load ramp-up and load ramp-down phases.

$$IAC = \int_0^t |e(t)|dt \tag{46}$$

$$ITAC = \int_0^t t|e(t)|dt \tag{47}$$

To further evaluate the control performance of the designed high-potential controller, Fig. 13 shows the error of the actual output voltage tracking the target voltage during the whole operation period. Meanwhile, the tracking performance shown in Fig. 13 is compared in detail using the integrated absolute cost (IAC) and the integrated absolute time

Table 8 – Quantitative comparison of control effects.

Stage	Controllers	Evaluation indicators	Deviation(b)	Tuning time
		Deviation(a)		
21s	SFF + FSPID	0	−0.5 mV	2s
Step-down	SFF + SPID	1 mV	−1mV	2.5s
	SFF	2 mV	−0.5 mV	4s
35s	SFF + FSPID	−2mV	0	0.5s
Step-up	SFF + SPID	−5.5 mV	1.6 mV	1.5s
	SFF	−6mV	2.8 mV	2.5s

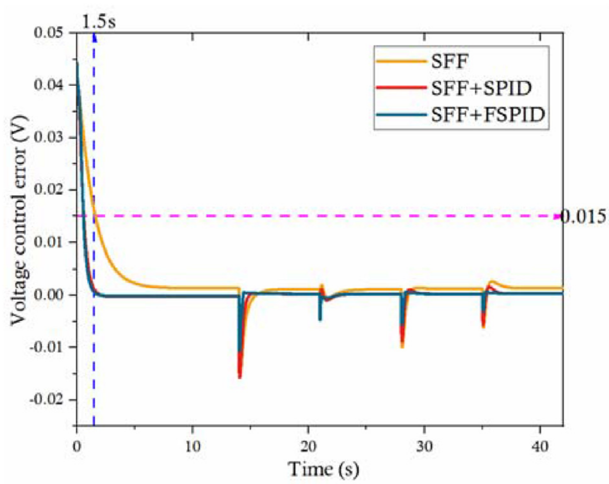


Fig. 13 — The tracking error analysis of high potential dynamic control.

Table 9 — Trajectory tracking performance comparison of different high-potential controllers.

Values		Evaluation indicators	ITAC
		IAC	
Controllers	SFF+FSPID	0.036	0.344
	SFF+SPID	0.043	0.464
	SFF	0.120	1.352

cost (ITAC) as shown in equation (46) and (47). The quantitative results are summarized in Table 9 and clearly show the low cost characteristics of the designed SFF + FSPID controller. IAC and ITAC are significantly reduced, which indicates that the voltage follows the desired value more accurately during the whole operation period.

Investigation of the self-humidification effect of cathode recirculation

Since the cathode recirculation is to re-feed the low oxygen content and high humidity exhaust from the stack outlet to the stack inlet, the high potential suppression based on oxygen partial pressure regulation will be accompanied by the humidification function. This humidification effect is an improvement to the current state-of-the-art self-humidifying fuel cell systems that suffer from water scarcity under low load conditions. Current automotive self-humidifying fuel cell systems have removed external humidifiers due to the pursuit of specific power and cost, and have instead achieved self-humidifying effects through structural material updates, hydrogen circulation systems, and counter-flow configurations. This solution strongly relies on the water production of the fuel cell, however, the low water production of the fuel cell under low load conditions can easily cause the membrane to dry out, thus reducing the performance and durability. Although the cathode recirculation in this paper is centered on high potential control based on oxygen partial pressure regulation, humidification still has a positive effect as an accompanying phenomenon. In the performance evaluation of the high-potential controller, the SFF+FSPID controller showed the most superior performance. Therefore, Fig. 14 shows the humidification performance when the optimal high-potential controller is implemented, and the PEMFC system is operated with the current profile shown in Fig. 10. Fig. 14(a) shows the relative humidity of the fuel cell cathode before and after the implementation of the high-potential controller, which is estimated by the model to directly evaluate the membrane humidity state but is not measurable. The results show that the fuel cell system without the high-potential controller is not fully humidified during low-load operation, which exposes the proton membrane to a water shortage environment. The system implementing the optimal high potential controller operates in a fully humidified environment except for the controller start-up phase, which means that the high potential suppression is satisfied while significantly increasing the cathode relative humidity. This has positive significance to solve the two major problems (high potential and low humidity) that cause durability decay under low load conditions of self-humidifying systems. In general, the relative humidity index inside the cathode of automotive fuel cell systems is unobservable due to the lack of essential in-situ sensors; therefore, the inlet air humidity at the stack inlet is widely used to measure the humidification effect. Fig. 14(b) compares the relative humidity at the stack inlet

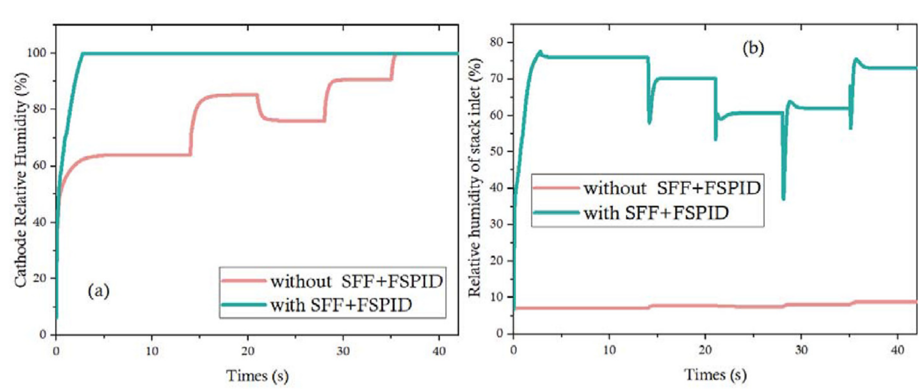


Fig. 14 — Self-humidification effect of configuring SFF+FSPID high-potential control strategy. (a) Relative humidity of the cell cathode, (b) Relative humidity of the inlet air at stack inlet.

before and after applying high-potential control, and this result has more practical application value. It can be seen that the humidity at the inlet of the stack without the optimal high-potential controller applied is below 10% for the whole operation period, which is an extremely terrible humidification effect. However, after the implementation of the optimal high-potential controller, the relative humidity is maintained at 60%–80% and increased by about 70% on average, which is a satisfactory humidification effect.

Conclusions

In this work, a comprehensive dynamic control model of an automotive self-humidifying fuel cell system with cathode recirculation is developed and the steady-state and transient performance of the model is validated by experimental data. The air compressor model is developed and validated using RBF neural network with a fitting accuracy of 99.85%; the cathode recirculation pump model based on BP neural network is developed and the error is controlled within ±5%; the steady-state prediction error of output voltage is within 1% and the dynamic error is within 3.2%. The developed model has the superior ability to accurately simulate the gas supply process and output performance. Then, based on the developed model, the dynamic and control problems of cathode recirculation in automotive high-power self-humidifying fuel cell systems under low load conditions are innovatively discussed to enhance durability. The optimal high potential control strategy based on oxygen partial pressure regulation is developed and evaluated through a series of low variable load conditions using cathode recirculation. The results show that the designed dynamic fuzzy logic PID controller with feedforward compensation (SFF + FSPID) has the best oxygen partial pressure control performance. The optimal high-potential control performance is achieved by controlling the oxygen partial pressure. The SFF + FSPID controller shows superior high-potential control performance in terms of tuning time, steady-state error, overshoot, and robustness. From the trajectory tracking ability of the whole operation condition, the IAC and ITAC of SFF + FSPID are the smallest, which are 0.036 and 0.334, respectively. The inlet relative humidity of the cathode is increased by 70% and the estimated humidity of the membrane is 100% while obtaining the best high potential control performance. Overall, the proposed dynamic fuzzy logic PID controller with feedforward compensation achieves accurate and fast control of high potentials under low load conditions and improves the humidification level, which has positive theoretical significance and application value for the development and implementation of the durability optimization control technology in current vehicle high-power self-humidifying fuel cell systems.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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REFERENCES

[1] Yang B, Li J, Li Y, Guo Z, Zeng K, Shu H, et al. A critical survey of proton exchange membrane fuel cell system control: summaries, advances, and perspectives. Int J Hydrogen Energy 2022;47(17):9986–10020.
[2] Liu Z, Zhang B, Xu S. Research on air mass flow-pressure combined control and dynamic performance of fuel cell system for vehicles application. Appl Energy 2022:309.
[3] Cano ZP, Banham D, Ye S, Hintennach A, Lu J, Fowler M, et al. Batteries and fuel cells for emerging electric vehicle markets. Nat Energy 2018;3(4):279–89.
[4] Wang X, Lu Y, Zhang B, Liu J, Xu S. Experimental analysis of an ejector for anode recirculation in a 10 kW polymer electrolyte membrane fuel cell system. Int J Hydrogen Energy 2022;47(3):1925–39.
[5] Wang J. System integration, durability and reliability of fuel cells: challenges and solutions. Appl Energy 2017;189:460–79.
[6] Zhao J, Li X. A review of polymer electrolyte membrane fuel cell durability for vehicular applications: degradation modes and experimental techniques. Energy Conversion and Management; 2019. p. 199.
[7] Yuan X-Z, Li H, Zhang S, Martin J, Wang H. A review of polymer electrolyte membrane fuel cell durability test protocols. J Power Sources 2011;196(22):9107–16.
[8] Ren P, Pei P, Li Y, Wu Z, Chen D, Huang S. Degradation mechanisms of proton exchange membrane fuel cell under typical automotive operating conditions. Prog Energy Combust Sci 2020:80.
[9] Tian T, Tang J, Guo W, Pan M. Accelerated life-time test of MEA durability under vehicle operating conditions in PEM fuel cell. Front Energy 2017;11(3):326–33.
[10] Zhang B, Wang X, Gong D, Xu S. Experimental analysis of the performance of the air supply system in a 120 kW polymer electrolyte membrane fuel cell system. Int J Hydrogen Energy 2022;47(50):21417–34.
[11] Chen H, Zhao X, Zhang T, Pei P. The reactant starvation of the proton exchange membrane fuel cells for vehicular applications: a review. Energy Convers Manag 2019;182:282–98.
[12] Daud WRW, Rosli RE, Majlan EH, Hamid SAA, Mohamed R, Husaini T. PEM fuel cell system control: a review. Renew Energy 2017;113:620–38.
[13] Shen Q, Hou M, Liang D, Zhou Z, Li X, Shao Z, et al. Study on the processes of start-up and shutdown in proton exchange membrane fuel cells. J Power Sources 2009;189(2):1114–9.
[14] Meier JC, Galeano C, Katsounaros I, Topalov AA, Kostka A, Schüth F, et al. Degradation mechanisms of Pt/C fuel cell catalysts under simulated start–stop conditions. ACS Catal 2012;2(5):832–43.
[15] Uno M, Tanaka K. Pt/C catalyst degradation in proton exchange membrane fuel cells due to high-frequency potential cycling induced by switching power converters. J Power Sources 2011;196(23):9884–9.
[16] Yu Y, Li H, Wang H, Yuan X-Z, Wang G, Pan M. A review on performance degradation of proton exchange membrane fuel cells during startup and shutdown processes: causes, consequences, and mitigation strategies. J Power Sources 2012;205:10–23.

[17] Panha K, Fowler M, Yuan X-Z, Wang H. Accelerated durability testing via reactants relative humidity cycling on PEM fuel cells. Appl Energy 2012;93:90–7.

[18] Seo D, Lee J, Park S, Rhee J, Choi SW, Shul Y-G. Investigation of MEA degradation in PEM fuel cell by on/off cyclic operation under different humid conditions. Int J Hydrogen Energy 2011;36(2):1828–36.

[19] Hasegawa T, Imanishi H, Nada M, Ikogi Y. Development of the fuel cell system in the mirai FCV. SAE Technical Paper Series; 2016.

[20] Pei P, Chen H. Main factors affecting the lifetime of Proton Exchange Membrane fuel cells in vehicle applications: a review. Appl Energy 2014;125:60–75.

[21] Rodosik S, Poirot-Crouvezier JP, Bultel Y. Impact of humidification by cathode exhaust gases recirculation on a PEMFC system for automotive applications. Int J Hydrogen Energy 2019;44(25):12802–17.

[22] Zhang Q, Tong Z, Tong S, Cheng Z. Self-humidifying effect of air self-circulation system for proton exchange membrane fuel cell engines. Renew Energy 2021;164:1143–55.

[23] Wilson MP, Yadha V, Reiser CA. Low power control of fuel cell open circuit voltage, EP2351134; 2011-08-03.

[24] Nathaniel Ian JOOS, Forte Paolo. System and method for controlling voltage of a fuel cell using a recirculation line. CA2913376A1. 2014. p. 11–27.

[25] Cheng S, Li J, Xu L, Ouyang M. Air supply system model with exhaust gas recirculation for improving the life of fuel cell. In: 2014 IEEE conference and expo transportation electrification asia-pacific (ITEC asia-pacific); 2014. p. 1–6. https://doi.org/10.1109/ITEC-AP.2014.6941250.

[26] Jiang H, Xu L, Fang C, Zhao X, Hu Z, Li J, et al. Experimental study on dual recirculation of polymer electrolyte membrane fuel cell. Int J Hydrogen Energy 2017;42(29):18551–9.

[27] Zhao X, Xu L, Fang C, Jiang H, Li J, Ouyang M. Study on voltage clamping and self-humidification effects of pem fuel cell system with dual recirculation based on orthogonal test method. Int J Hydrogen Energy 2018;43(33):16268–78.

[28] Kim BJ, Kim MS. Studies on the cathode humidification by exhaust gas recirculation for PEM fuel cell. Int J Hydrogen Energy 2012;37(5):4290–9.

[29] Shao Y, Xu L, Zhao X, Li J, Hu Z, Fang C, et al. Comparison of self-humidification effect on polymer electrolyte membrane fuel cell with anodic and cathodic exhaust gas recirculation. Int J Hydrogen Energy 2020;45(4):3108–22.

[30] Xu L, Fang C, Hu J, Cheng S, Li J, Ouyang M, et al. Self-humidification of a polymer electrolyte membrane fuel cell system with cathodic exhaust gas recirculation. J Electrochem Energy Convers Storage 2018;15(2).

[31] Zhang Q, Tong Z, Tong S. Effect of cathode recirculation on high potential limitation and self-humidification of hydrogen fuel cell system. J Power Sources 2020:468.

[32] Pukrushpan JT, Stefanopoulou AG, Peng H. Control of fuel cell breathing. IEEE Control Syst Mag 2004;24(2):30–46.

[33] Chen H, Liu Z, Ye X, Yi L, Xu S, Zhang T. Air flow and pressure optimization for air supply in proton exchange membrane fuel cell system. Energy 2022;238.

[34] Wang Y, Li H, Feng H, Han K, He S, Gao M. Simulation study on the PEMFC oxygen starvation based on the coupling algorithm of model predictive control and PID. Energy Conversion and Management; 2021. p. 249.

[35] Pukrushpan JT, Peng H, Stefanopoulou AG. Control-oriented modeling and analysis for automotive fuel cell systems. J Dynamic Syst Measure Cont-Trans ASME 2004;126(1):14–25.

[36] Sankar K, Jana AK. Nonlinear multivariable sliding mode control of a reversible PEM fuel cell integrated system. Energy Convers Manag 2018;171:541–65.

[37] Chen X, Xu J, Liu Q, Chen Y, Wang X, Li W, et al. Active disturbance rejection control strategy applied to cathode humidity control in PEMFC system. Energy Conversion and Management; 2020. p. 224.

[38] Zhao D, Xia L, Dang H, Wu Z, Li H. Design and control of air supply system for PEMFC UAV based on dynamic decoupling strategy. Energy Convers Manag 2022:253.

[39] Baroud Z, Benmiloud M, Benalia A, Ocampo-Martinez C. Novel hybrid fuzzy-PID control scheme for air supply in PEM fuel-cell-based systems. Int J Hydrogen Energy 2017;42(15):10435–47.

[40] Beirami H, Shabestari AZ, Zerafat MM. Optimal PID plus fuzzy controller design for a PEM fuel cell air feed system using the self-adaptive differential evolution algorithm. Int J Hydrogen Energy 2015;40(30):9422–34.

[41] Yuan H, Dai H, Ming P, Zhan J, Wang X, Wei X. A fuzzy extend state observer-based cascade decoupling controller of air supply for vehicular fuel cell system. Energy Convers Manag 2021:236.

[42] Benchouia NE, Derghal A, Mahmah B, Madi B, Khochemane L, Hadjadj Aoul E. An adaptive fuzzy logic controller (AFLC) for PEMFC fuel cell. Int J Hydrogen Energy 2015;40(39):13806–19.

[43] Liu Z, Chen H, Peng L, Ye X, Xu S, Zhang T. Feedforward-decoupled closed-loop fuzzy proportion-integral-derivative control of air supply system of proton exchange membrane fuel cell. Energy 2022:240.

[44] Ou K, Wang Y-X, Li Z-Z, Shen Y-D, Xuan D-J. Feedforward fuzzy-PID control for air flow regulation of PEM fuel cell system. Int J Hydrogen Energy 2015;40(35):11686–95.

[45] Li Q, Chen W, Liu Z, Li M, Ma L. Development of energy management system based on a power sharing strategy for a fuel cell-battery-supercapacitor hybrid tramway. J Power Sources 2015;279:267–80.

[46] Wang Y-X, Qin F-F, Ou K, Kim Y-B. Temperature control for a polymer electrolyte membrane fuel cell by using fuzzy rule. IEEE Trans Energy Convers 2016;31(2):667–75.

[47] Qi Y, Espinoza-Andaluz M, Thern M, Li T, Andersson M. Dynamic modelling and controlling strategy of polymer electrolyte fuel cells. Int J Hydrogen Energy 2020;45(54):29718–29.